SYNCRIP-related Neurodevelopmental Disorder: A Comprehensive Research Report

Target disease: SYNCRIP-related Neurodevelopmental Disorder **MONDO ID:** MONDO:0800456 **Causal gene: SYNCRIP** (Synaptotagmin-binding Cytoplasmic RNA-Interacting Protein; also known as **hnRNP-Q**; HGNC:16918; NCBI Gene 10492; Ensembl ENSG00000135316; UniProt O60506; chromosome 6q14.3)

Summary

SYNCRIP-related Neurodevelopmental Disorder (SYNCRIP-NDD) is a rare, autosomal-dominant Mendelian condition caused by heterozygous, typically *de novo*, variants in the gene SYNCRIP, which encodes the RNA-binding protein hnRNP-Q. The disorder was established when a large-scale intellectual-disability trio meta-analysis flagged SYNCRIP as a statistically supported candidate gene, and was confirmed by subsequent case reports describing additional individuals with *de novo* loss-of-function and RNA-recognition-motif (RRM) missense variants. The core clinical picture comprises intellectual disability/developmental delay, frequently with autism spectrum features, and variable epilepsy (including myoclonic-atonic seizures) and structural brain malformations such as periventricular nodular heterotopia and widened subarachnoid spaces.

The molecular mechanism is **haploinsufficiency of hnRNP-Q**. SYNCRIP is one of the most loss-of-function-intolerant genes in the human genome (gnomAD pLI = 1.00, LOEUF = 0.228), and pathogenic variants are predominantly truncating (frameshift/nonsense) alleles that reduce functional protein dosage, alongside rarer missense variants in the conserved RRM2 domain that impair RNA binding. Although SYNCRIP is ubiquitously expressed across tissues, the disease is CNS-selective, implying a neuron-specific dependence on precise hnRNP-Q dosage rather than tissue-restricted expression. Mechanistically, hnRNP-Q controls cortical neurogenesis by maintaining the temporal transcription-factor program of radial glial cells through a **phase-separation-dependent** recruitment of mRNA-stabilization complexes that

regulates Notch signaling. Pathogenic human mutations weaken this phase-separation capacity, stalling the transcriptional progression of neural progenitors and producing cortical layering defects and learning/memory impairment in mouse models.

SYNCRIP-NDD belongs to the broader family of **"hnRNPopathies"** — neurodevelopmental disorders caused by haploinsufficiency of heterogeneous nuclear ribonucleoprotein genes (e.g., *HNRNPU*, *HNRNPK*) — which converge on dysregulation of post-transcriptional RNA metabolism (splicing, mRNA stability, transport, and local translation) in neurons. The disorder is managed symptomatically; there is no disease-modifying therapy. This report synthesizes evidence from human clinical genetics, population constraint databases, model-organism studies (mouse and *Drosophila*), and in vitro molecular biology.

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1. Disease Information

Overview. SYNCRIP-related Neurodevelopmental Disorder is a monogenic neurodevelopmental disorder characterized primarily by intellectual disability/global developmental delay, with a high frequency of autism spectrum disorder features and a variable occurrence of epilepsy and brain malformations. It results from heterozygous pathogenic variants in *SYNCRIP*, a gene encoding the RNA-binding protein hnRNP-Q, a member of the heterogeneous nuclear ribonucleoprotein (hnRNP) family that regulates mRNA splicing, stability, transport, and translation.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0800456
Gene (HGNC)	SYNCRIP, HGNC:16918
NCBI Gene	10492
Ensembl	ENSG00000135316
UniProt	O60506 (HNRPQ_HUMAN)
Cytogenetic locus	6q14.3 (GRCh38 chr6:85,607,779–85,644,063)
OMIM	Not recorded for the disease (no dedicated phenotype MIM at time of research)
ICD-10/ICD-11	No disease-specific code; coded under generic neurodevelopmental/intellectual-disability categories
MeSH	No disease-specific MeSH; indexed under Intellectual Disability / Neurodevelopmental Disorders

Synonyms / alternative names: SYNCRIP-related intellectual disability; hnRNP-Q-related neurodevelopmental disorder; (protein-based) Synaptotagmin-binding cytoplasmic RNA-interacting protein-related disorder.

Source of information. The entry is derived from **aggregated disease-level resources and small published case series**, not EHR/individual-patient data. The evidence base is small — a handful of published individuals plus a de novo mutation meta-analysis — supplemented by extensive functional data from model organisms and in vitro studies. Monarch Initiative confirms the MONDO definition but shows sparse phenotype annotation, consistent with an ultra-rare, recently delineated disorder.

2. Etiology

Primary cause — genetic. SYNCRIP-NDD is caused by heterozygous pathogenic variants in *SYNCRIP*. These arise **predominantly *de novo***, consistent with a dominant mechanism acting through haploinsufficiency. Two mutational classes are documented:

1. **Loss-of-function (truncating) variants** — frameshift and nonsense variants producing prematurely terminated/unstable transcript, reducing functional hnRNP-Q dosage (haploinsufficiency).
2. **Missense variants in the RNA-recognition-motif (RRM) domains** — notably in the conserved RRM2 domain — impairing RNA binding.

Semino et al. (2021) reported three individuals: two with frameshift variants (the more severely affected) and one with an RRM2 missense variant: *"Two frameshift variants in the more severely affected individuals, likely result in haploinsufficiency. The third missense variant lies in the conserved RNA recognition motif (RRM) 2 domain likely affecting RNA-binding"* PMID: 34157790.

SYNCRIP was first nominated as an intellectual-disability gene by a meta-analysis of 2,104 trios: *"Statistical analyses identified 10 new candidate ID genes: DLG4, PPM1D, RAC1, SMAD6, SON, SOX5, SYNCRIP, TCF20, TLK2 and TRIP12"* PMID: 27479843.

Genetic risk factors. The causal variants themselves are the risk factors. There are no reported susceptibility loci or validated modifier genes. The gene's extreme constraint (Section 4) means any functional haploinsufficient lesion carries high disease risk.

Environmental risk factors. None established. As a monogenic *de novo* disorder, advanced parental age is a general (non-specific) risk factor for *de novo* mutation burden, not specifically quantified for SYNCRIP.

Protective factors. None identified (genetic or environmental).

Gene–environment interactions. None documented.

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3. Phenotypes

The phenotypic spectrum is derived from a small published cohort (principally the 3 individuals of Semino et al. plus the meta-analysis probands). Reported features, with suggested HPO terms:

Phenotype	Type	HPO term	Frequency (small cohort)	Onset	Severity/ course
Intellectual disability / global developmental delay	Cognitive	HP:0001249 / HP:0001263	~100%	Infancy/ childhood	Variable (mild–severe); stable, lifelong
Autism spectrum disorder / autistic features	Behavioral	HP:0000729	~2/3	Early childhood	Variable
Epilepsy (incl. myoclonic-atonic seizures)	Neurological sign	HP:0001250 / HP:0002123	~1/3	Childhood	Variable; episodic seizures
Periventricular nodular heterotopia	Neuroradiological/ structural	HP:0032388 / HP:0007165	Reported	Congenital	Static malformation
Widened subarachnoid space	Neuroradiological	HP:0011283	Reported	Congenital/ infancy	Static
Speech/ language delay	Cognitive/ communicative	HP:0000750	Common	Childhood	Lifelong

Phenotype characteristics. Onset is in the **neonatal-to-childhood** window (developmental). Severity is **variable**, correlating loosely with variant class — frameshift/truncating (haploinsufficiency) variants were seen in the more severely affected individuals, whereas the RRM2 missense case was reportedly milder. The disorder is **non-progressive/static** (a developmental encephalopathy with fixed deficits rather than neurodegeneration), though epilepsy introduces an episodic component.

Quality-of-life impact. Intellectual disability and ASD features substantially impair adaptive functioning, communication, and independence, typically requiring lifelong educational and caregiver support. Epilepsy adds seizure-related morbidity and treatment burden. Formal QoL instrument data (EQ-5D, SF-36, PROMIS) are not available for this ultra-rare disorder.

> **Note on uncertainty:** Frequencies are based on <10 published individuals and are provisional. Larger cohorts (e.g., via GeneMatcher) are needed for robust penetrance/frequency figures.

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4. Genetic / Molecular Information

Causal gene. *SYNCRIP* (HGNC:16918; NCBI Gene 10492; Ensembl ENSG00000135316), chromosome 6q14.3. Encodes hnRNP-Q / Synaptotagmin-binding cytoplasmic RNA-interacting protein (UniProt O60506).

Protein architecture. hnRNP-Q contains **three RNA-recognition-motif (RRM) domains**, a **unique N-terminal acidic domain (AcD)**, and a **C-terminal RGG box**. The acidic domain is a non-canonical, structurally distinct nucleic-acid-binding region: *"The acid-rich sequence segment is unique to SYNCRIP and the closely related protein hnRNP R"* PMID: 27081926. The AcD binds ATP at physiological concentrations despite lacking an RRM fold PMID: 31791586.

Pathogenic variants (ClinVar, transcript NM_006372.5). NCBI ClinVar returns 193 SYNCRIP records: 121 Uncertain significance, 19 Pathogenic, 4 Likely pathogenic, 1 Pathogenic/Likely pathogenic, 17 Likely benign, 1 Benign. Pathogenic small variants are **predominantly truncating**:

Variant (NM_006372.5)	Protein	Class	Classification
c.1473dup	p.Gln492fs	Frameshift	Pathogenic
c.700_701del	p.Ile234fs	Frameshift	Pathogenic
c.1336C>T	p.Arg446Ter	Nonsense	Pathogenic
c.438dup	p.Pro147fs	Frameshift	Likely pathogenic
c.858_859del	p.Gly287fs	Frameshift	Pathogenic

Additional pathogenic entries are **large 6q14 copy-number deletions/duplications** spanning *SYNCRIP* (e.g., GRCh38 6q14.3-15 chr6:85,370,716–90,669,793 ×1). Most missense SNVs remain **variants of uncertain significance (VUS)**.

Variant classification (ACMG/AMP). Truncating variants in a highly LoF-constrained gene meet strong pathogenicity criteria (e.g., PVS1 + de novo PS2/PM6). RRM-domain missense variants require functional evidence (RNA-binding/phase-separation assays) to be upgraded from VUS.

Allele frequency. Pathogenic variants are absent/vanishingly rare in population databases (gnomAD), consistent with *de novo*, highly penetrant disease alleles.

Somatic vs germline. Germline (constitutional); *de novo* in affected probands.

Functional consequences. Loss of function / haploinsufficiency is the dominant mechanism. Frameshift/nonsense alleles reduce functional dosage; RRM2 missense alleles impair RNA binding, with a demonstrated phase-separation-weakening effect (Section 6).

gnomAD constraint (v2.1.1). SYNCRIP is among the most LoF-intolerant genes:

Metric	Value	Interpretation
pLI	1.00 (0.99999999986)	Extreme LoF intolerance
LOEUF (oe_lof upper)	0.228	Most-constrained decile (<0.35)
observed/expected LoF	0.131 (9 obs vs 68.9 exp)	~87% depletion of LoF variants
LoF Z	6.12	Strong constraint
Missense Z	5.48 (oe_mis 0.537)	Significant missense constraint
Synonymous Z	−0.15	Neutral (validates model)

These support a **haploinsufficiency** disease mechanism (dosage sensitivity).

Expression (GTEx v8). SYNCRIP is **ubiquitously expressed** (median 33.9 TPM across 54 tissues; range 6.2–89.9 TPM), highest in cultured fibroblasts (89.9), EBV-transformed lymphocytes (88.8), and thyroid (63.6); brain regions 11–28 TPM (cerebellar hemisphere 27.5, frontal cortex BA9 16.2, hippocampus 13.0, cortex 12.7); lowest in whole blood (6.2). Ubiquitous expression contrasted with CNS-selective disease implies **neuron-specific dosage dependence**.

Modifier genes / epigenetics. None established. The paralog *HNRNPR* (hnRNP-R) shares high sequence homology and partially overlapping functions; in *Drosophila* both correspond to the single ortholog *Syncrip* (Syp) PMID: 30214903. Whether hnRNP-R dosage modifies disease severity is untested. No episignature/methylation marker is known.

Chromosomal abnormalities. 6q14.3 deletions/duplications encompassing *SYNCRIP* are documented as pathogenic CNVs in ClinVar.

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5. Environmental Information

No environmental, lifestyle, or infectious contributors are known for SYNCRIP-NDD. It is a monogenic *de novo* genetic disorder. Environmental factors (e.g., parental-age effects on *de novo* mutation rate) are generic to all *de novo* disorders and not specific to this condition. **Not applicable / none documented.**

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **de novo variant in *SYNCRIP*** arises — either a frameshift/nonsense allele or an RRM-domain missense allele.
2. **Truncating alleles lead to** reduced functional hnRNP-Q protein (haploinsufficiency); **missense RRM2 alleles lead to** impaired RNA binding; **both branches converge** on insufficient functional hnRNP-Q activity in neural cells.
3. Reduced/impaired hnRNP-Q **results in weakened liquid-liquid phase-separation (LLPS) capacity** — pathogenic human mutations fail to form stable biomolecular condensates (demonstrated in vitro and in mouse) PMID: 39776340.
4. Weakened phase separation **leads to** failure to recruit mRNA-stabilization complexes that maintain transcription of **temporal transcription factors** in radial glial cells (RGCs).
5. Loss of temporal-transcription-factor maintenance **dysregulates Notch signaling** governing RGC fate, which **stalls the transcriptional progression** of neural progenitors through their temporal identity program (inferred causal linkage from mouse knockout data).

6. Stalled progenitor temporal dynamics **result in** disrupted late-stage neurogenesis, abnormal neuronal migration/localization, and **superficial cortical-layer defects**.
7. Abnormal cortical architecture and connectivity **lead to** the clinical phenotypes: intellectual disability, autism spectrum features, epilepsy, and structural malformations (periventricular nodular heterotopia) — plus learning/memory impairment demonstrated in mouse models.

Branch note: In addition to the neurogenesis branch (steps 3–6), hnRNP-Q has **post-mitotic neuronal roles** (synaptic plasticity, neurite morphogenesis, local translation) that likely contribute to the cognitive/behavioral phenotype independently of cortical layering, though their specific contribution in patients is **inferred, not demonstrated**.

Supporting detail

Molecular pathways. The central pathway is **Notch signaling** regulating RGC temporal identity, downstream of hnRNP-Q-mediated mRNA stabilization. Wu et al. (2025) showed: *"Syncrip knockout in RGCs disrupts differentiation and abnormal neuronal localization, ultimately resulting in superficial cortical layer defects as well as learning and memory impairments in mice"* and *"pathogenic human mutations in Syncrip weaken its phase-separation capability, failing to form stable complexes normally"* PMID: 39776340.

Cellular processes. Regulation of neural-progenitor temporal identity; neuronal differentiation and migration; synaptic plasticity; neurite/dendrite morphogenesis. In mammalian neurons, hnRNP-Q1 binds mRNAs encoding the actin-nucleation machinery: *"hnRNP Q1 bound to a set of identified mRNAs encoding the components of the actin nucleation-promoting Cdc42/N-WASP/Arp2/3 complex"* PMID: 22493061.

Protein dysfunction. Loss of function via haploinsufficiency (truncating) or impaired RNA binding (RRM2 missense), both reducing functional ribonucleoprotein condensates (weakened LLPS). The unique acidic domain mediates non-canonical nucleic-acid/ATP binding relevant to condensate behavior PMID: 27081926; PMID: 31791586.

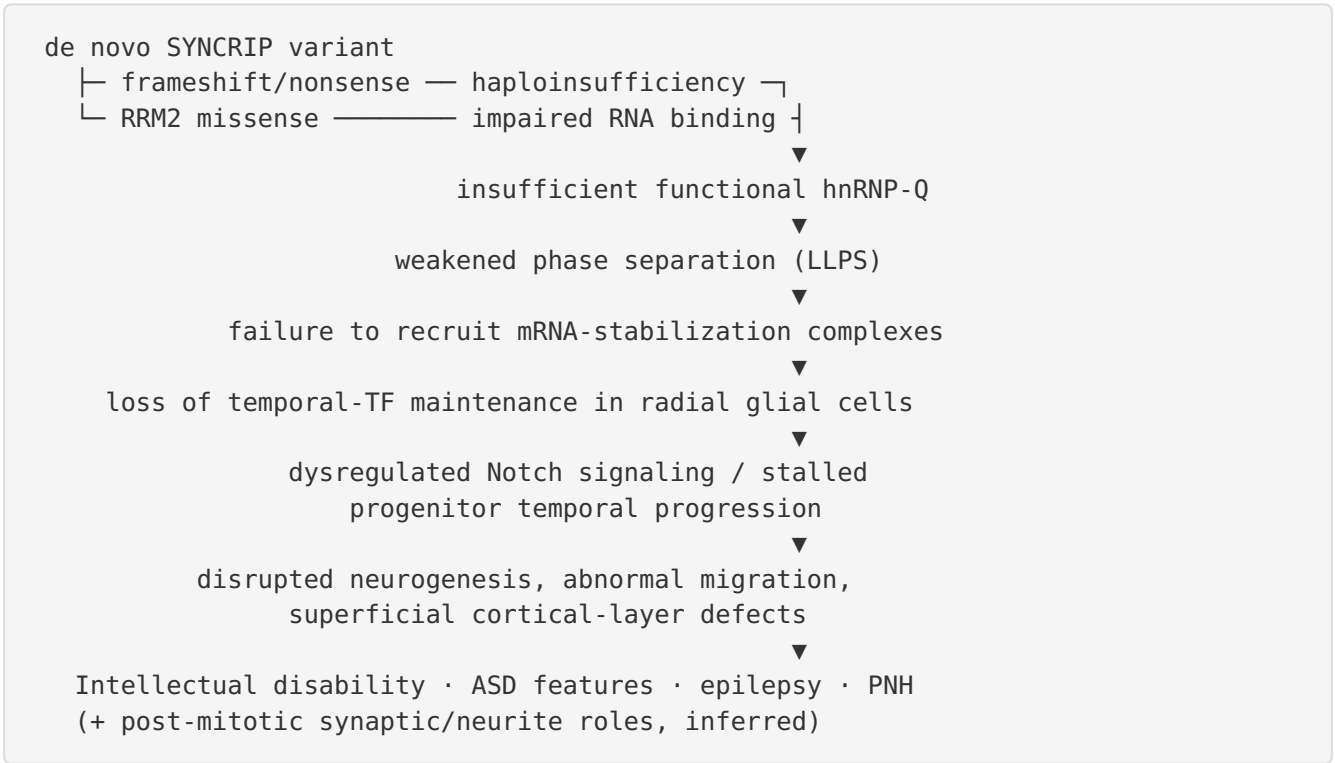
Molecular functions (QuickGO; UniProt O60506, 62 annotations). - *Molecular function:* RNA binding (GO:0003723), mRNA 5′-UTR binding (GO:0048027), nucleic acid binding (GO:0003676), protein binding (GO:0005515). - *Biological process:* negative regulation of translation (GO:0017148), positive regulation of cytoplasmic translation (GO:2000767), CRD-mediated mRNA stabilization (GO:0070934), negative regulation of deadenylation-dependent mRNA decay (GO:1900152), mRNA splicing via spliceosome (GO:0000398), RNA processing (GO:0006396). - *Cellular component:* nucleus (GO:0005634), cytoplasm/cytosol (GO:0005737/

GO:0005829), ribonucleoprotein complex (GO:1990904), CRD-mediated mRNA stability complex (GO:0070937/GO:0106002), GAIT complex (GO:0097452), catalytic step 2 spliceosome (GO:0071013).

Evolutionary conservation (Drosophila). The single fly ortholog *Syncrip/Syp* controls neuroblast temporal identity via the **Imp→Syp switch** PMID: 28394252; PMID: 38031099, and is required for activity-dependent synaptic growth: *"activity-dependent accumulation of Msp300 in the postsynaptic compartment of the Drosophila larval neuromuscular junction is regulated by the conserved RNA binding protein Syncrip/hnRNP Q"* PMID: 32040548.

Suggested GO/CL terms. Biological processes: GO:0021895 (cerebral cortex neuron differentiation), GO:0007219 (Notch signaling pathway), GO:0070934 (CRD-mediated mRNA stabilization), GO:0017148 (negative regulation of translation). Cell types: CL:0000679 (glutamatergic neuron), CL:0000047 (neural stem cell), CL:0000031 (neuroblast), CL:0000129 (radial glial cell / apical radial glia).

Mechanistic diagram



7. Anatomical Structures Affected

Organ level. Primary organ: the **brain** (central nervous system). The disorder is CNS-selective despite ubiquitous SYNCRIP expression. No consistent primary involvement of other organ systems.

Body systems. Nervous system (UBERON:0001016). Specifically, the **cerebral cortex** (UBERON:0000956) and its developing germinal zones; the **periventricular region** (site of nodular heterotopia); and the **subarachnoid space** (UBERON:0002360, widened).

Tissue and cell level. Nervous tissue. Key cell populations: **radial glial cells / apical progenitors** (CL:0000129), **neuroblasts/intermediate progenitors** (CL:0000031), and **cortical projection neurons** (CL:0000679, glutamatergic neurons). Post-mitotic neurons (including hippocampal neurons implicated in learning/memory in mouse models) are also affected through synaptic/morphogenic roles.

Subcellular level. hnRNP-Q localizes to the **nucleus** (GO:0005634) and **cytoplasm/cytosol** (GO:0005737/GO:0005829), functioning within **ribonucleoprotein complexes/granules** (GO:1990904), the **CRD-mediated mRNA stability complex** (GO:0070937), the **GAIT complex** (GO:0097452), and the **spliceosome** (GO:0071013). Biomolecular condensates (phase-separated granules) are the critical subcellular structures whose assembly is impaired.

Localization / lateralization. Cortical malformations (periventricular nodular heterotopia, widened subarachnoid spaces) are typically **bilateral**, consistent with a diffuse developmental mechanism.

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8. Temporal Development

Onset. Congenital/developmental. Cortical malformations originate during fetal corticogenesis; developmental delay manifests in infancy/early childhood. Onset pattern is **insidious/chronic** (a static developmental encephalopathy), not acute.

Progression. Cognitive/structural deficits are **static** (fixed developmental deficits) rather than neurodegenerative. There are no defined disease "stages." Epilepsy, when present, introduces an **episodic** seizure course. Overall, a **chronic, lifelong** condition.

Patterns / critical periods. The **critical period of vulnerability** is prenatal/perinatal corticogenesis, when radial-glial temporal programming and neuronal migration occur — the window in which SYNCRIP haploinsufficiency exerts its developmental effect. There is no spontaneous remission of the core neurodevelopmental phenotype; seizures may respond to anti-seizure medication.

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9. Inheritance and Population

Epidemiology. Prevalence and incidence are unknown/unestablished — the disorder is ultra-rare, with fewer than ~10 published individuals. No registry-based prevalence estimates exist. It contributes to the broad category of monogenic intellectual disability (collectively ~1% of the population, but each individual gene is rare).

Inheritance pattern. Autosomal dominant, essentially always **de novo** (sporadic). No confirmed vertical transmission reported.

Penetrance. Presumed **high/complete** for reported pathogenic truncating variants, though formal penetrance cannot be estimated from the tiny cohort.

Expressivity. Variable — severity correlates loosely with variant class (truncating/haploinsufficiency more severe than RRM2 missense).

Genetic anticipation. Not applicable (not a repeat-expansion disorder).

Germline mosaicism. Not specifically reported; theoretically possible as for other *de novo* dominant disorders (relevant to recurrence-risk counseling).

Founder effects / consanguinity. Not applicable — *de novo* dominant mechanism; no founder alleles; consanguinity not relevant.

Carrier frequency. Not applicable (dominant *de novo*). gnomAD shows depletion of LoF variants (9 observed vs 68.9 expected) — pathogenic alleles are not carried in the general population.

Population demographics. No ethnic/geographic predilection reported. Sex ratio not established (cohort too small). Age distribution: affected from birth/infancy; lifelong.

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10. Diagnostics

Recommended approach. Diagnosis is **molecular**, via broad genomic sequencing in a child with unexplained developmental delay/intellectual disability ± ASD ± epilepsy ± brain malformation.

Genetic testing (primary diagnostic modality). - **Trio whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** is the highest-yield approach, enabling *de novo* variant detection and classification. GeneMatcher-type data sharing is essential for ultra-rare genes. - **Intellectual disability / epilepsy / ASD gene panels** that include *SYNCRIP* can detect SNVs/indels. - **Chromosomal microarray (CMA)** detects 6q14.3 deletions/duplications encompassing *SYNCRIP*. - Single-gene testing has limited utility given phenotypic nonspecificity. - Karyotyping/FISH only for suspected large structural rearrangements; mtDNA and repeat-expansion testing are **not applicable**.

Variant interpretation. Apply ACMG/AMP criteria: truncating variants in this LoF-constrained gene (PVS1) plus *de novo* status (PS2/PM6) support pathogenicity; RRM-domain missense variants typically remain VUS pending functional assays (RNA binding, phase-separation).

Clinical tests / imaging. Brain MRI is informative — may reveal **periventricular nodular heterotopia** and **widened subarachnoid spaces**. **EEG** is indicated when seizures are suspected (may show epileptiform activity, including patterns consistent with myoclonic-atonic epilepsy). Routine labs, metabolic workup, and biopsies are typically normal/noncontributory (no specific biomarker exists).

Biomarkers / omics diagnostics. No validated protein, metabolite, or methylation biomarker (episignature) is established. RNA-sequencing for aberrant splicing is a research tool, not a clinical test.

Clinical criteria / differential diagnosis. No disorder-specific diagnostic criteria. Differential diagnosis includes other **hnRNPopathies** and monogenic ID/ASD/epilepsy syndromes — e.g., *HNRNPU*-related disorder (early-onset epilepsy, severe ID; PMID: 28393272), *HNRNPK*-related Au-Kline syndrome (distinctive facial gestalt; PMID: 29904177), *MED13L* haploinsufficiency (PMID: 28645799), and other heterotopia-associated genes (e.g., *FLNA*). Molecular testing distinguishes these.

Screening. No population newborn or carrier screening (ultra-rare, *de novo*). Cascade testing is generally unnecessary given *de novo* origin, but parental testing confirms de novo status for recurrence-risk counseling.

11. Outcome / Prognosis

Survival / mortality. No evidence of reduced life expectancy from the core disorder; it is **not directly life-limiting**. Mortality risk, if any, relates to seizure complications in severe epilepsy. No formal survival statistics exist.

Morbidity and function. The principal burden is **neurodevelopmental disability** — intellectual disability, communication/speech impairment, and ASD-related adaptive deficits — producing lifelong dependence of varying degree. Epilepsy adds morbidity in a subset.

Disease course. Chronic and **static** (non-degenerative). Core deficits persist; developmental gains occur with intervention but intellectual disability remains. Complications: seizures, feeding/behavioral issues, and those secondary to cortical malformation.

Recovery potential. No cure; **symptomatic improvement** with early intervention (therapies, special education) is possible but core cognitive impairment is permanent.

Prognostic factors. Variant class is the main apparent prognostic factor — truncating/haploinsufficiency variants were associated with more severe phenotypes than the RRM2 missense variant in the published series PMID: 34157790. Seizure control and severity of cortical malformation also influence outcome. No validated prognostic biomarkers.

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12. Treatment

There is no disease-modifying or targeted therapy. Management is **symptomatic and supportive**, following general neurodevelopmental-disorder best practice.

Pharmacotherapy. Anti-seizure medications for epilepsy, selected per seizure type (myoclonic-atonic epilepsy may respond to valproate, levetiracetam, lamotrigine, or specific regimens; choice individualized). Behavioral/ASD-associated symptoms managed per standard practice. No pharmacogenomic guidance specific to SYNCRIP. (NCIT: C15215 Anticonvulsant Therapy.)

Advanced therapeutics. None available. Gene, RNA-based, cell, targeted, or immunotherapies are **not applicable/not developed**. Haploinsufficiency is conceptually amenable to future dosage-restoring strategies (e.g., ASO-mediated upregulation or gene supplementation), but none exist for SYNCRIP.

Surgical/interventional. Epilepsy surgery only in rare refractory cases with a resectable focus; generally not applicable given diffuse malformation.

Supportive and rehabilitative (mainstay). - **Physical therapy** (NCIT: C15415) for hypotonia/motor delay. - **Occupational therapy** (NCIT: C15321). - **Speech-language therapy** (NCIT: C15402) for communication delay. - Special education, behavioral therapy for ASD features, and family/caregiver support.

Experimental treatments. No disease-specific clinical trials (no NCT identifiers) exist for SYNCRIP-NDD.

Treatment strategy. Multidisciplinary developmental-pediatrics/neurology care: seizure management, developmental therapies, educational support, and genetic counseling. **Personalized medicine** is currently limited to variant-informed prognosis and counseling.

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13. Prevention

Primary prevention. Not possible for *de novo* mutations. General preconception health applies but does not specifically prevent this disorder.

Secondary prevention / early detection. Early genetic diagnosis via WES/WGS enables **early intervention** (developmental therapies, seizure monitoring), improving functional outcomes though not altering the underlying genetic lesion.

Genetic screening / reproductive options. For families with an affected child (de novo), recurrence risk is low but non-zero (germline mosaicism). **Prenatal diagnosis** or **preimplantation genetic testing (PGT-M)** for the specific familial variant can be offered in subsequent pregnancies.

Counseling. Genetic counseling is central: explaining *de novo* dominant inheritance, low recurrence risk (with caveat of possible germline mosaicism), and reproductive options. Parental testing confirms de novo status.

Immunization / public health / environmental / prophylaxis. Not applicable.

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14. Other Species / Natural Disease

Taxonomy and orthologs. SYNCRIP is evolutionarily conserved. Orthologs include: - *Mus musculus* (NCBI Taxon:10090): *Syncrip* (used in the definitive mouse knockout study). - *Drosophila melanogaster* (NCBI Taxon:7227): single ortholog *Syncrip* (*Syp*), corresponding to both human paralogs hnRNP-Q and hnRNP-R PMID: 30214903.

Natural disease in other species. No naturally occurring SYNCRIP-related disease is documented in companion animals or wildlife (no OMIA entry noted). The animal involvement is **experimental** (model organisms), not natural disease.

Comparative biology. The temporal-identity function of Syp/hnRNP-Q is deeply conserved from flies to mammals: *Drosophila* uses the Imp→Syp gradient switch to program neuroblast temporal identity PMID: 28394252; PMID: 38031099, paralleling the mammalian radial-glial temporal program disrupted in disease. This conservation strengthens causal inference.

Transmission / zoonosis. Not applicable (genetic disorder).

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15. Model Organisms

Mouse (*Mus musculus*, MGI). The key model is a **conditional *Syncrip* knockout in radial glial cells**. Wu et al. (2025) showed this disrupts late-stage neurogenesis, stalls transcriptional progression, causes abnormal neuronal localization, superficial cortical-layer defects, and **learning/memory impairments** — recapitulating core features of the human disorder — and demonstrated that **human pathogenic mutations weaken phase separation** PMID: 39776340. This is the strongest model linking genotype to mechanism to phenotype.

Drosophila (*D. melanogaster*, FlyBase). The single ortholog *Syp* is a workhorse for dissecting neuroblast temporal identity (Imp→Syp switch) and activity-dependent synaptic plasticity (Msp300/Nesprin-1) PMID: 28394252; PMID: 32040548. It models progenitor-timing and synaptic functions but, as a single gene standing for two human paralogs, does not distinguish hnRNP-Q from hnRNP-R.

Cellular / in vitro models. Human neuroblastoma (SH-SY5Y) knockdown of hnRNP-Q/hnRNP-R revealed distinct downstream gene programs affecting neurodegeneration and inflammation pathways PMID: 30214903. Rodent hippocampal neurons demonstrated hnRNP-Q control of cytoskeletal (Cdc42/N-WASP/Arp2/3) mRNAs and neurite morphogenesis PMID: 22493061 and

translational regulation under stress PMID: 34944075. In vitro biophysics characterized the unique acidic domain and its ATP/nucleic-acid binding relevant to phase separation PMID: 27081926; PMID: 31791586.

Model characteristics. The mouse conditional knockout **recapitulates** cortical layering defects and cognitive impairment. **Limitations:** models use complete loss (knockout) rather than the human heterozygous haploinsufficient state; the *Drosophila* single ortholog conflates paralog-specific functions; patient-derived iPSC/organoid models of specific variants are not yet reported. iPSC-derived cortical organoids carrying patient RRM2 missense or truncating variants would be a valuable next-generation model.

Evidence Base

PMID	Title (abbrev.)	Evidence type	How it supports findings
27479843	*Meta-analysis of 2,104 trios... 10 new ID genes*	Human genetics	Statistically nominates SYNCRIP as an ID gene (F001)
34157790	*Further evidence for de novo SYNCRIP variants*	Human clinical	Defines two variant classes; core phenotypes (F001)
39776340	*Posttranscriptional control of neural progenitors by Syncrip*	Mouse / in vitro	Phase-separation mechanism; Notch/temporal-TF program; human mutations weaken LLPS (F002)
27081926	*Acidic domain is a unique feature of SYNCRIP*	In vitro / structural	Protein architecture; unique acidic domain (F003)
32040548	*Syncrip required for activity-induced Msp300*	Drosophila	Conserved synaptic-plasticity role (F003)
22493061	*hnRNP Q regulates Cdc42-mediated morphogenesis*	Neuronal in vitro	Cytoskeletal mRNA regulation; neurite morphogenesis (F003)
28394252	*Steroid-hormone induction of temporal gene expression*	Drosophila	Imp→Syp temporal switch (F003)
38031099	*Imp in INPs and newborn neurons*	Drosophila	Imp/Syp gradients in progenitor timing (F003)
30214903	*Gene expression profiles controlled by hnRNP Q and R*	In vitro	Paralog relationship; distinct downstream programs (F003, F008)
31791586	*ATP binds nucleic-acid domains beyond RRM*	Biophysics	Acidic-domain ATP binding relevant to LLPS (F003)
28393272	*Heterozygous HNRNPU variants cause epilepsy + ID*	Human clinical	Comparator hnRNPopathy (F008)
39976380	*Expanding HNRNPU-related disorder spectrum*	Human clinical	Comparator hnRNPopathy phenotype (F008)
29904177	*Au-Kline syndrome (HNRNPK)*	Human clinical	Comparator hnRNPopathy (F008)
41325101	*RNArchitects: how hnRNPs shape neuronal landscapes*	Review	hnRNP dysfunction as convergent NDD mechanism (F008)

PMID	Title (abbrev.)	Evidence type	How it supports findings
32865115	*RNA-binding proteins in neurological disease*	Review	RBP dysregulation in NDD (F008)

Supporting database evidence (non-PMID): gnomAD v2.1.1 constraint (pLI=1.00, LOEUF=0.228) for haploinsufficiency (F004); NCBI ClinVar (193 records; truncating-variant predominance) for variant spectrum (F005); GTEx v8 for ubiquitous expression vs CNS-selective disease (F006); QuickGO/UniProt O60506 for molecular functions (F007); Monarch Initiative for MONDO definition and sparse annotation.

Limitations and Knowledge Gaps

1. **Tiny clinical cohort.** Fewer than ~10 published individuals; phenotype frequencies, penetrance, expressivity, and prognosis are provisional. No natural-history or registry data.
2. **No OMIM/ICD/Orphanet-specific identifiers** at the time of research, reflecting recent delineation and under-annotation (confirmed sparse in Monarch).
3. **Genotype–phenotype correlation is preliminary** — the truncating-more-severe-than-missense pattern rests on 3 individuals.
4. **Variant interpretation gap.** Most SYNCRIP missense variants are VUS (121/193 ClinVar records); functional assays are not routinely available clinically.
5. **Mechanistic models use complete knockout**, not the human heterozygous state; paralog redundancy (hnRNP-R) and its possible modifier role are untested in the disease context.
6. **No biomarkers, no episignature, no targeted therapy, and no disease-specific clinical trials.**
7. **No patient-derived iPSC/organoid models** of specific variants reported.

Proposed Follow-up Experiments / Actions

1. **Cohort expansion via GeneMatcher/matchmaking** to assemble ≥20–30 individuals, enabling robust phenotype-frequency, penetrance, and genotype–phenotype analyses, and to seek an OMIM phenotype entry.
2. **Functional assays for VUS reclassification** — systematic RNA-binding (CLIP/EMSA) and phase-separation (in vitro droplet, cellular condensate) assays for recurrent missense variants.
3. **Patient-specific iPSC-derived cortical organoids** carrying truncating vs RRM2 missense variants, to test whether human heterozygous dosage reproduces the temporal-TF/Notch/layering defects seen in mouse knockouts and to quantify condensate deficits.
4. **Heterozygous (Syncrip$^{+/-}$) and knock-in missense mouse models** to better mimic the human haploinsufficient state and to test paralog (hnRNP-R) modifier effects.
5. **Transcriptome/splicing profiling (RNA-seq)** of patient cells to search for a diagnostic aberrant-splicing or expression signature (potential biomarker/episignature analog).
6. **Mechanism-guided therapeutic exploration** — because the mechanism is haploinsufficiency, evaluate dosage-restoring strategies (e.g., ASO-mediated upregulation of the wild-type allele) in cellular/mouse models.
7. **Standardized natural-history data collection** (development, seizures, neuroimaging) to define prognosis and critical intervention windows.

Consensus Answer

SYNCRIP-related neurodevelopmental disorder (MONDO:0800456) is a rare, autosomal-dominant, typically *de novo* Mendelian condition caused by heterozygous loss-of-function (frameshift/nonsense → haploinsufficiency) or RNA-recognition-motif missense variants in **SYNCRIP** (hnRNP-Q), presenting with intellectual disability, frequent autism spectrum features, and variable epilepsy and cortical malformation (e.g., periventricular nodular heterotopia). Mechanistically, insufficient hnRNP-Q impairs RNA binding and phase separation, disrupting the Notch-dependent temporal transcription-factor program by which radial glial cells build the cortex, yielding cortical layering defects and lifelong cognitive impairment; management is symptomatic with no disease-modifying therapy.